

Week-6-PP2

September 6, 2026

1 Interactive Scatter Plot with Plotly

1.1 Week 6 Practice Program

1.1.1 Problem Statement

Create an interactive scatter plot showing the relationship between house size and price.

1.1.2 Dataset

The dataset contains house size in square feet and corresponding house price in lakhs.

```
[1]: import pandas as pd
import plotly.express as px
```

```
[2]: data = {
    "House Size (sq.ft.)": [
        800,
        1000,
        1200,
        1400,
        1600,
        1800,
        2000,
        2200,
        2500,
        2800
    ],
    "House Price (Lakhs)": [
        35,
        45,
        55,
        65,
        78,
        90,
        105,
        120,
        140,
```

```
        165
    ]
}
```

```
[3]: df = pd.DataFrame(data)
```

```
df
```

```
[3]:
```

	House Size (sq.ft.)	House Price (Lakhs)
0	800	35
1	1000	45
2	1200	55
3	1400	65
4	1600	78
5	1800	90
6	2000	105
7	2200	120
8	2500	140
9	2800	165

```
[4]: print("Dataset Shape:", df.shape)

print("\nHouse Sizes:")
print(df["House Size (sq.ft.)"].tolist())

print("\nHouse Prices:")
print(df["House Price (Lakhs)"].tolist())
```

Dataset Shape: (10, 2)

House Sizes:

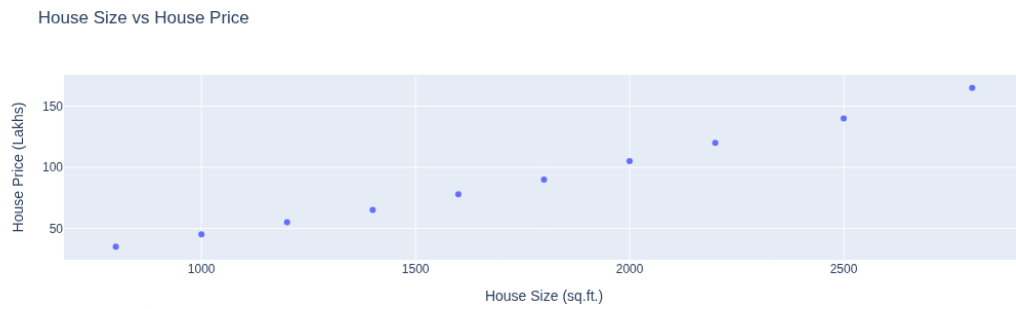
[800, 1000, 1200, 1400, 1600, 1800, 2000, 2200, 2500, 2800]

House Prices:

[35, 45, 55, 65, 78, 90, 105, 120, 140, 165]

```
[5]: fig = px.scatter(
    df,
    x="House Size (sq.ft.)",
    y="House Price (Lakhs)",
    title="House Size vs House Price"
)

fig.show()
```

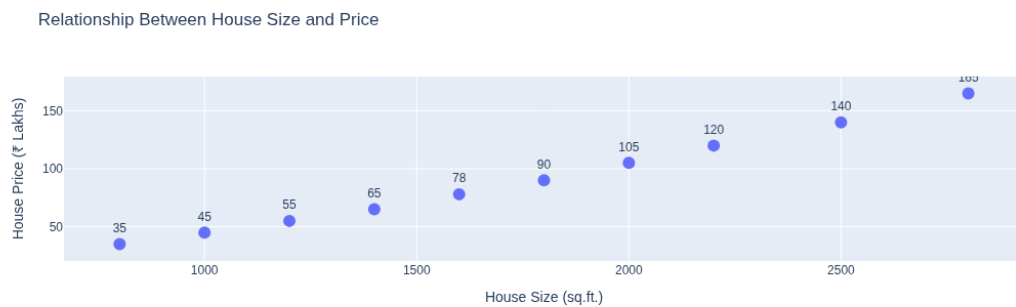


```
[6]: fig = px.scatter(
    df,
    x="House Size (sq.ft.)",
    y="House Price (Lakhs)",
    title="Relationship Between House Size and Price",
    text="House Price (Lakhs)"
)

fig.update_traces(
    marker=dict(size=12),
    textposition="top center"
)

fig.update_layout(
    xaxis_title="House Size (sq.ft.)",
    yaxis_title="House Price ( Lakhs)"
)

fig.show()
```



```
[7]: sorted_df = df.sort_values(
      by="House Size (sq.ft.)",
      ascending=True
    )

sorted_df
```

```
[7]:
```

	House Size (sq.ft.)	House Price (Lakhs)
0	800	35
1	1000	45
2	1200	55
3	1400	65
4	1600	78
5	1800	90
6	2000	105
7	2200	120
8	2500	140
9	2800	165

```
[8]: highest = df.loc[
      df["House Price (Lakhs)"].idxmax()
    ]

print("Most Expensive House:")
print(highest)
```

```
Most Expensive House:
House Size (sq.ft.)    2800
House Price (Lakhs)     165
Name: 9, dtype: int64
```

```
[9]: lowest = df.loc[
      df["House Price (Lakhs)"].idxmin()
    ]

print(
    "Least Expensive House:",
    lowest["House Size (sq.ft.)"],
    "sq.ft. - ",
    lowest["House Price (Lakhs)"],
    "Lakhs"
)
```

```
Least Expensive House: 800 sq.ft. - 35 Lakhs
```

```
[10]: average_price = df["House Price (Lakhs)"].mean()

print(
```

```
    "Average House Price:",  
    round(average_price, 2),  
    "Lakhs"  
)
```

Average House Price: 89.8 Lakhs

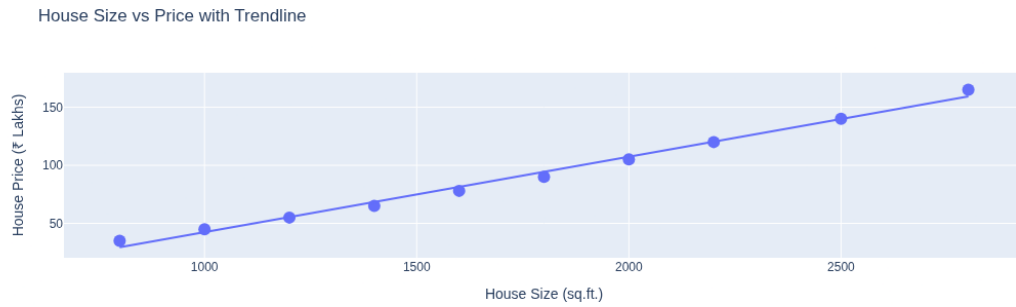
```
[11]: average_size = df["House Size (sq.ft.)"].mean()  
  
print(  
    "Average House Size:",  
    round(average_size, 2),  
    "sq.ft."  
)
```

Average House Size: 1730.0 sq.ft.

```
[12]: correlation = df[  
    "House Size (sq.ft.)"  
].corr(  
    df["House Price (Lakhs)"]  
)  
  
print(  
    "Correlation between House Size and Price:",  
    round(correlation, 3)  
)
```

Correlation between House Size and Price: 0.996

```
[14]: fig = px.scatter(  
    df,  
    x="House Size (sq.ft.)",  
    y="House Price (Lakhs)",  
    title="House Size vs Price with Trendline",  
    trendline="ols"  
)  
  
fig.update_traces(  
    marker=dict(size=12)  
)  
  
fig.update_layout(  
    xaxis_title="House Size (sq.ft.)",  
    yaxis_title="House Price ( Lakhs)"  
)  
  
fig.show()
```



```
[15]: print("HOUSE PRICE ANALYSIS")
print("-" * 40)

print("Total Houses:", len(df))

print(
    "Smallest House:",
    df["House Size (sq.ft.)"].min(),
    "sq.ft."
)

print(
    "Largest House:",
    df["House Size (sq.ft.)"].max(),
    "sq.ft."
)

print(
    "Lowest Price: ",
    df["House Price (Lakhs)"].min(),
    "Lakhs"
)

print(
    "Highest Price: ",
    df["House Price (Lakhs)"].max(),
    "Lakhs"
)

print(
    "Average Price: ",
    round(df["House Price (Lakhs)"].mean(), 2),
    "Lakhs"
```

```
)  
  
print(  
    "Correlation:",  
    round(correlation, 3)  
)
```

HOUSE PRICE ANALYSIS

```
-----  
Total Houses: 10  
Smallest House: 800 sq.ft.  
Largest House: 2800 sq.ft.  
Lowest Price: 35 Lakhs  
Highest Price: 165 Lakhs  
Average Price: 89.8 Lakhs  
Correlation: 0.996
```

```
[16]: fig.write_html(  
        "house_size_price_scatter.html"  
    )  
  
print("Interactive scatter plot saved successfully.")
```

Interactive scatter plot saved successfully.

```
[17]: df.to_csv(  
        "house_price_data.csv",  
        index=False  
    )  
  
print("Dataset saved successfully.")
```

Dataset saved successfully.

```
[ ]:
```